

# Preprocessing

19/2/2024

## Description

This file enables preprocessing of upper-paleolithic sign sequences as published at <https://www.signbase.org>. The csv file with latest objects and sign codings can be downloaded from there.

## Session Info

Give the session info (reduced).

```
## [1] "R version 3.6.3 (2020-02-29)"
```

```
## [1] "x86_64-pc-linux-gnu"
```

## Load libraries

If the libraries are not installed yet, you need to install them using, for example, the command: `install.packages("ggplot2")`.

```
#library(dplyr)
```

## Preprocessing

Preprocessing and storage of cleaned file for signs. Load data from github repo and remove missing rows.

```
# load data from github repo
sign.data <- read.csv(
  "~/Github/UpperPalCombinatorics/data/signBase/Sb2Artifact-2024-02-20.csv")
# remove certain columns
sign.data <- subset(sign.data, select = -c(id, comments, short_description, references))
# remove rows for which there is no sign coding
sign.data <- sign.data[!sign.data$sign_coding == "", ]
```

Remove UTF8 characters which do not represent actual signs on the objects, i.e “punctuation”.

```
# remove some special characters
coding.clean <- gsub('[!?', '', sign.data$sign_coding)
# remove sign strings which are not discrete, i.e. overlapping sign types
# -> indicated by parentheses '()'
coding.clean <- gsub("\\\\([^()]*\\\\)", "", coding.clean)
# remove sign strings which are not in linear order, i.e. those grouped in visual fields
# -> indicated by square brackets '[]'
coding.clean <- gsub("\\\\([.*?]*\\\\)", "", coding.clean)
```

```

# replace multiple underscores by single underscore
coding.clean <- gsub("( _ )+", " _ ", coding.clean)
# trim the underscores and white spaces from beginning and end of sequences
coding.clean <- trimws(coding.clean, which = "both", whitespace = "[ \\_ ]")

# add column to original data frame again
sign.data$coding_clean <- coding.clean
# again remove rows for which there is no sign coding (after cleaning)
sign.data <- sign.data[!sign.data$coding_clean == "", ]

```

Safe to file.

```

write.csv(sign.data,
  file = "~/Github/UpperPalCombinatorics/data/signBase/signBase_clean.csv",
  row.names = F)

```